Technology Stack

Date	10 JUNE 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements. Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice. **Technology Stack Details**

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	Streamlit	Streamlit is selected because it enables the fast, efficient construction of data-driven user interfaces directly within Python, allowing seamless visualization of complex media elements like voice waveforms and real-time score indicators.

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	Python (`app.py`, `scoring_engine.py`)	Python serves as the project foundation due to its mature ecosystems for digital signal processing, advanced machine learning integration, and robust data manipulation capabilities.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	File System Storage (`/uploads/`, `/reports/`)	A structured local file directory storage system provides fast performance, simplicity, and low overhead for staging temporary raw user recordings and storing compiled evaluation PDFs without database complexities.

4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	Docker / Streamlit Community Cloud	Docker ensures reliable containerized deployment consistency across environments, isolating essential system frameworks and heavy model layer dependencies for smooth production deployment.

5	**Version Control & CI/CD**	GitHub	GitHub provides source code control management and supports clear version isolation across feature branches during modular development cycles.
	(e.g., GitHub, GitLab, Jenkins)		

6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	OpenAI Whisper, Sentence-BERT, Librosa, ReportLab	**Whisper** provides highly accurate speech-to-text conversion; **SentenceBERT** calculates semantic text mapping; **Librosa** extracts core signal arrays (energy, pause metrics); and **ReportLab** automates clean PDF summary reports.